TrES-5b

R-1333

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_141023` · created 2026-07-31T20:47:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456953.6624 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.045 +/- 0.037
Transit depth (Rp/Rs)^2	0.21 +/- 0.34 [%]
Orbital Inclination (inc)	79.85 +/- 3.07
Airmass coefficient 1 (a1)	0.938 +/- 0.023
Airmass coefficient 2 (a2)	0.054 +/- 0.022
Scatter in the residuals of the lightcurve fit is	2.5 %
Best Comparison Star	#2 - [399, 72]
Optimal Aperture	3.43
Optimal Annulus	5.7
Transit Duration (day)	0.022 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.045 ± 0.037 vs 0.142 (deviation 2.62σ)
Duration fit vs published	0.022 d vs 0.0758 d (deviation 1.99σ)
Mid-time O–C	-0.3 min
Depth SNR	1.2
Residual scatter	2.5 %

Light curve

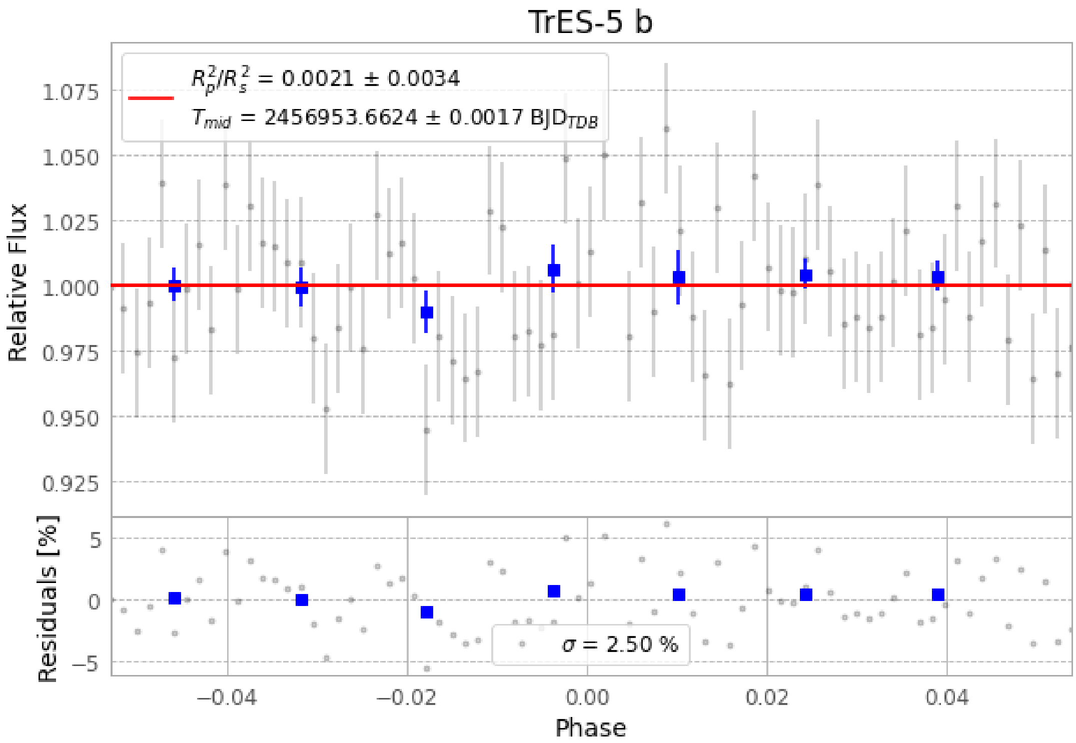

Figure 1 — FinalLightCurve_TrES-5 b_2014-10-22.png (SHA-256 ecb43c009bc23b85…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [311, 313], 2 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 7049a7f73fbcd164…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

77 FITS frames (51 MB), `TRES-5141023015712.FITS` → `TRES-5141023054512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-141023.log (973808879ced…), FinalLightCurve_TrES-5 b_2014-10-22.png (ecb43c009bc2…), FinalLightCurve_TrES-5 b_2014-10-22.csv (d356c600b155…)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)